RELIABLE RENTALS

Logical Database Design Report

Elkin Huertas, Yasemin Yılmaz, Jennifer Rogers

Table of Contents

1. Introduction

Scope

- Derivation of relations and foreign keys from the conceptual model
- Relational schema notation for all relations
- Normalisation validation to Third Normal Form (3NF)
- Validation against five representative user transactions
- Definition of all integrity constraints
- Logical ER diagram with foreign keys as attributes

Design Assumptions

The following assumptions underpin the logical design. All assumptions from the conceptual model are carried forward and supplemented with logical-level clarifications.

#	Assumption
A1	A vehicle is located at exactly one outlet at any given time. OutletNumber is stored as a foreign key in VEHICLE.
A2	A hire agreement is associated with exactly one vehicle and exactly one client. Both RegistrationNumber and ClientNumber are stored as foreign keys in HIRE_AGREEMENT.
A3	A client may exist in the system with zero hire agreements (e.g. a newly registered client who has not yet hired a vehicle).
A4	A vehicle may exist in the system with zero hire agreements (e.g. a newly acquired vehicle not yet hired).
A5	An outlet may temporarily have zero vehicles and zero staff (e.g. a new outlet being set up).
A6	ClientNumber is the primary key for CLIENT. DrivingLicenseNumber is retained as an alternate (candidate) key and must be unique.
A7	Redundant client and vehicle data referenced in the case study hire agreement description (client name, address, phone, vehicle model, make) is not stored as direct attributes of HIRE_AGREEMENT. This data is accessible via foreign key joins and its inclusion would violate normalisation principles.
A8	Each staff member is assigned to exactly one outlet. OutletNumber is stored as a foreign key in STAFF.
A9	Name attributes for clients and staff are stored as two separate attributes: FirstName and LastName.
A10	Address attributes are stored as single composite string attributes at the logical level.
A11	DailyHireRate is stored as a decimal value representing the cost in the local currency per day.
A12	MileageAfter in HIRE_AGREEMENT may be NULL if the vehicle has not yet been returned at the time of recording.

2. Derive Relations and Foreign Keys

Mapping Rules Applied

- Each strong entity becomes a relation; its primary key is carried forward.
- Each one-to-many relationship introduces a foreign key on the 'many' side.
- No many-to-many relationships exist in this model; no junction tables are required beyond HIRE_AGREEMENT.
- Redundant derived attributes (client name, vehicle make/model on hire agreement) are omitted per assumption A7.

Foreign Key Derivation Summary

Relationship	Multiplicity	FK Placed In	FK Attribute	References
OUTLET stocks VEHICLE	1 to 0..*	VEHICLE	OutletNumber	OUTLET(OutletNumber)
OUTLET employs STAFF	1 to 0..*	STAFF	OutletNumber	OUTLET(OutletNumber)
CLIENT makes HIRE_AGREEMENT	1 to 0..*	HIRE_AGREEMENT	ClientNumber	CLIENT(ClientNumber)
VEHICLE covered by HIRE_AGREEMENT	1 to 0..*	HIRE_AGREEMENT	RegistrationNumber	VEHICLE(RegistrationNumber)

Relational Schema

OUTLET

```
OUTLET(__OutletNumber__, Address, PhoneNumber, FaxNumber)
```

VEHICLE

```
VEHICLE(__RegistrationNumber__, Model, Make, EngineSize, Capacity,
        CurrentMileage, DailyHireRate, OutletNumber*)
```

CLIENT

```
CLIENT(__ClientNumber__, FirstName, LastName, HomeAddress,
       PhoneNumber, DateOfBirth, DrivingLicenseNumber(AK))
```

HIRE_AGREEMENT

```
HIRE_AGREEMENT(__HireNumber__, StartDate, EndDate,
               MileageBefore, MileageAfter,
               ClientNumber*, RegistrationNumber*)
```

STAFF

```
STAFF(__StaffNumber__, FirstName, LastName, HomeAddress,
      HomePhoneNumber, DateOfBirth, Sex, DateJoined,
      JobTitle, Salary, OutletNumber*)
```

Detailed Attribute Tables

OUTLET

Attribute	Data Type	Key	Constraints	Description
OutletNumber	VARCHAR(10)	PK	NOT NULL, UNIQUE	Unique identifier for each outlet
Address	VARCHAR(200)		NOT NULL	Full street address of the outlet
PhoneNumber	VARCHAR(20)		NOT NULL	Contact phone number
FaxNumber	VARCHAR(20)		NOT NULL	Fax number for the outlet

VEHICLE

Attribute	Data Type	Key	Constraints	Description
RegistrationNumber	VARCHAR(15)	PK	NOT NULL, UNIQUE	Unique vehicle registration plate
Model	VARCHAR(50)		NOT NULL	Vehicle model name
Make	VARCHAR(50)		NOT NULL	Vehicle manufacturer/brand
EngineSize	VARCHAR(10)		NOT NULL	Engine displacement
Capacity	INT		NOT NULL, ≥ 1	Passenger seating capacity
CurrentMileage	INT		NOT NULL, ≥ 0	Odometer reading at last update
DailyHireRate	DECIMAL(8,2)		NOT NULL, > 0	Cost per day to hire the vehicle
OutletNumber	VARCHAR(10)	FK	NOT NULL	Current outlet location — ref. OUTLET(OutletNumber)

CLIENT

Attribute	Data Type	Key	Constraints	Description
ClientNumber	VARCHAR(10)	PK	NOT NULL, UNIQUE	Unique system-generated client ID
FirstName	VARCHAR(50)		NOT NULL	Client's first name
LastName	VARCHAR(50)		NOT NULL	Client's last name
HomeAddress	VARCHAR(200)		NOT NULL	Client's residential address
PhoneNumber	VARCHAR(20)		NOT NULL	Client's contact phone number
DateOfBirth	DATE		NOT NULL	Client's date of birth

DrivingLicenseNumber	VARCHAR(20)	AK	NOT NULL, UNIQUE	Government-issued driving licence number

HIRE_AGREEMENT

Attribute	Data Type	Key	Constraints	Description
HireNumber	VARCHAR(10)	PK	NOT NULL, UNIQUE	Unique identifier for each hire agreement
StartDate	DATE		NOT NULL	Date hire period commences
EndDate	DATE		NOT NULL	Planned date of vehicle return
MileageBefore	INT		NOT NULL, ≥ 0	Odometer reading at start of hire
MileageAfter	INT		≥ 0, nullable	Odometer reading at end of hire (NULL if not yet returned)
ClientNumber	VARCHAR(10)	FK	NOT NULL	Client who made the agreement — ref. CLIENT(ClientNumber)
RegistrationNumber	VARCHAR(15)	FK	NOT NULL	Vehicle covered — ref. VEHICLE(RegistrationNumber)

STAFF

Attribute	Data Type	Key	Constraints	Description
StaffNumber	VARCHAR(10)	PK	NOT NULL, UNIQUE	Unique identifier for each staff member
FirstName	VARCHAR(50)		NOT NULL	Staff member's first name
LastName	VARCHAR(50)		NOT NULL	Staff member's last name
HomeAddress	VARCHAR(200)		NOT NULL	Staff member's residential address
HomePhoneNumber	VARCHAR(20)		NOT NULL	Staff member's personal phone number
DateOfBirth	DATE		NOT NULL	Staff member's date of birth
Sex	CHAR(1)		NOT NULL, IN (M,F,O)	Staff member's sex as recorded
DateJoined	DATE		NOT NULL	Date the staff member joined the company
JobTitle	VARCHAR(100)		NOT NULL	Staff member's job role or title
Salary	DECIMAL(10,2)		NOT NULL, > 0	Staff member's annual salary
OutletNumber	VARCHAR(10)	FK	NOT NULL	Outlet where staff member is based — ref. OUTLET(OutletNumber)

3. Validate Logical Model Using Normalization to 3NF

3.1 First Normal Form (1NF)

A relation is in 1NF if every attribute contains only atomic (indivisible) values and there are no repeating groups.

Relation	1NF	Justification
OUTLET	Yes	All attributes are atomic. OutletNumber is a single identifier. No repeating groups.
VEHICLE	Yes	All attributes are atomic. EngineSize is stored as a single string. No multi-valued attributes.
CLIENT	Yes	Name is split into FirstName and LastName (atomic). No repeating groups.
HIRE_AGREEMENT	Yes	All attributes are atomic. No multi-valued attributes. MileageAfter is a single nullable integer.
STAFF	Yes	Name is split into FirstName and LastName. All other attributes are atomic. No repeating groups.

3.2 Second Normal Form (2NF)

A relation is in 2NF if it is in 1NF and every non-key attribute is fully functionally dependent on the entire primary key (relevant only to relations with composite primary keys).

Relation	2NF	Justification
OUTLET	Yes	Single-attribute PK (OutletNumber). 2NF is automatically satisfied — no partial dependencies possible.
VEHICLE	Yes	Single-attribute PK (RegistrationNumber). All non-key attributes depend fully on the registration number.
CLIENT	Yes	Single-attribute PK (ClientNumber). All non-key attributes depend fully on the client number.
HIRE_AGREEMENT	Yes	Single-attribute PK (HireNumber). All non-key attributes — including the two FKs — depend fully on the hire number. No composite key exists, so no partial dependencies are possible.
STAFF	Yes	Single-attribute PK (StaffNumber). All non-key attributes depend fully on the staff number.

3.3 Third Normal Form (3NF)

A relation is in 3NF if it is in 2NF and there are no transitive dependencies — every non-key attribute must depend directly on the primary key, not on another non-key attribute.

Relation	3NF?	Justification
OUTLET	Yes	No transitive dependencies. Address, PhoneNumber, and FaxNumber all depend solely on OutletNumber.

VEHICLE	Yes	No transitive dependencies. OutletNumber (FK) depends on RegistrationNumber (the vehicle's current location), not on any other non-key attribute. Model and Make are independent attributes describing the vehicle, not derived from each other.
CLIENT	Yes	No transitive dependencies. DrivingLicenseNumber is an alternate key, not a non-key attribute that determines anything else. All attributes depend solely on ClientNumber.
HIRE_AGREEMENT	Yes	No transitive dependencies. StartDate, EndDate, MileageBefore, and MileageAfter all depend directly on HireNumber. ClientNumber and RegistrationNumber are foreign keys that depend on HireNumber, not on each other.
STAFF	Yes	No transitive dependencies. Salary and JobTitle both depend on StaffNumber (the individual employee), not on each other. OutletNumber (FK) depends on StaffNumber, not on another non-key attribute.

4. Validate Logical Model Against User Transactions

Transaction 1 — List all vehicles currently stocked at a given outlet

Query (Plain English)

"What vehicles are currently available at the Edinburgh outlet, and what are their daily hire rates?"

Entities Involved

- OUTLET
- VEHICLE

Steps

1. Start in the OUTLET relation and locate the record where OutletNumber matches the Edinburgh outlet's identifier.
2. Move to the VEHICLE relation and retrieve all records where the OutletNumber foreign key matches the Edinburgh outlet number identified in step 1.
3. From those VEHICLE records, read the RegistrationNumber, Model, Make, and DailyHireRate attributes.

This transaction confirms that the FK OutletNumber in VEHICLE correctly links vehicles to their current outlet, enabling a straightforward one-to-many traversal from OUTLET to VEHICLE.

Transaction 2 — Retrieve full details of a client's hire history

Query (Plain English)

"What hire agreements has client C0042 entered into, and which vehicles were involved in each agreement?"

Entities Involved

- CLIENT
- HIRE_AGREEMENT
- VEHICLE

Steps

4. Start in the CLIENT relation and locate the record where ClientNumber = 'C0042'. Confirm the client exists and retrieve their name.
5. Move to the HIRE_AGREEMENT relation and retrieve all records where the ClientNumber foreign key matches 'C0042'.
6. For each matching hire agreement, read the HireNumber, StartDate, EndDate, MileageBefore, and MileageAfter attributes.
7. For each hire agreement, take the RegistrationNumber foreign key and use it to look up the corresponding record in the VEHICLE relation to retrieve the vehicle's Model and Make.

This transaction confirms that the FK ClientNumber in HIRE_AGREEMENT links hire agreements to clients, and the FK RegistrationNumber links each agreement to the specific vehicle hired. It validates a three-entity traversal: CLIENT → HIRE_AGREEMENT → VEHICLE.

Transaction 3 — Find all staff employed at a specific outlet

Query (Plain English)

"Which staff members are based at the Glasgow outlet, and what are their job titles and salaries?"

Entities Involved

- OUTLET
- STAFF

Steps

8. Start in the OUTLET relation and identify the record where the outlet's address or name corresponds to Glasgow. Note the OutletNumber.
9. Move to the STAFF relation and retrieve all records where the OutletNumber foreign key matches the Glasgow outlet number.
10. From those STAFF records, read the FirstName, LastName, JobTitle, and Salary attributes.

This transaction validates the FK OutletNumber in STAFF, confirming the one-to-many relationship between OUTLET and STAFF is navigable in both directions.

Transaction 4 — Calculate the mileage driven during a specific hire agreement

Query (Plain English)

"How many miles were driven during hire agreement H0117, and which client and vehicle does this agreement involve?"

Entities Involved

- HIRE_AGREEMENT
- CLIENT
- VEHICLE

Steps

11. Start in the HIRE_AGREEMENT relation and locate the record where HireNumber = 'H0117'.
12. Calculate the mileage driven by subtracting MileageBefore from MileageAfter. If MileageAfter is NULL, the vehicle has not yet been returned and the calculation cannot be completed.
13. Take the ClientNumber foreign key from the hire agreement record and look it up in the CLIENT relation to retrieve the client's FirstName, LastName, and PhoneNumber.
14. Take the RegistrationNumber foreign key from the hire agreement record and look it up in the VEHICLE relation to retrieve the vehicle's Model, Make, and EngineSize.

This transaction validates that HIRE_AGREEMENT holds sufficient data to calculate mileage directly, and that foreign keys provide navigable links to both CLIENT and VEHICLE for supplementary details.

Transaction 5 — Find all active hire agreements for vehicles at a given outlet

Query (Plain English)

"Which vehicles from the Aberdeen outlet are currently out on hire, and who are the clients that hired them?"

Entities Involved

- OUTLET
- VEHICLE
- HIRE_AGREEMENT
- CLIENT

Steps

15. Start in the OUTLET relation and locate the record corresponding to Aberdeen. Note the OutletNumber.
16. Move to the VEHICLE relation and retrieve all records where OutletNumber matches the Aberdeen outlet.
17. For each vehicle retrieved, take its RegistrationNumber and search the HIRE_AGREEMENT relation for records where the RegistrationNumber FK matches and the EndDate is on or after today's date (or MileageAfter is NULL), indicating the agreement is still active.
18. For each active hire agreement found, take the ClientNumber FK and look it up in the CLIENT relation to retrieve the client's FirstName, LastName, and PhoneNumber.

This transaction is the most complex of the five, requiring a four-entity traversal: OUTLET → VEHICLE → HIRE_AGREEMENT → CLIENT. It validates that all foreign key linkages function correctly end-to-end and that the schema can support operational queries spanning the full breadth of the data model.

5. Integrity Constraints

5.1 Primary Key Constraints

Each relation must have a primary key that uniquely identifies every tuple. The primary key must be NOT NULL and UNIQUE.

Relation	Primary Key	Type	Notes
OUTLET	OutletNumber	Single attribute	Cannot be null or duplicated.
VEHICLE	RegistrationNumber	Single attribute	Government-issued plate number. Globally unique. Cannot be null.
CLIENT	ClientNumber	Single attribute	Cannot be null or duplicated.
HIRE_AGREEMENT	HireNumber	Single attribute	Cannot be null or duplicated.
STAFF	StaffNumber	Single attribute	Cannot be null or duplicated.

5.2 Referential Integrity / Foreign Key Constraints

Foreign key constraints ensure that every FK value in a relation corresponds to an existing primary key value in the referenced relation. Orphaned records are not permitted.

FK Attribute	Defined In	References	On Delete	On Update
OutletNumber	VEHICLE	OUTLET(OutletNumber)	RESTRICT	CASCADE
OutletNumber	STAFF	OUTLET(OutletNumber)	RESTRICT	CASCADE
ClientNumber	HIRE_AGREEMENT	CLIENT(ClientNumber)	RESTRICT	CASCADE
RegistrationNumber	HIRE_AGREEMENT	VEHICLE(RegistrationNumber)	RESTRICT	CASCADE

- RESTRICT on Delete: Prevents deletion of an OUTLET that still has vehicles or staff assigned to it. Prevents deletion of a CLIENT or VEHICLE that is referenced by existing hire agreements.
- CASCADE on Update: If a primary key value changes (e.g. OutletNumber is renumbered), the change propagates automatically to all referencing foreign keys.

5.3 Alternate Key Constraints

An alternate key (AK) is a candidate key that was not selected as the primary key. It must still enforce uniqueness and is typically implemented via a UNIQUE constraint.

Relation	Alternate Key	Constraint	Justification
CLIENT	DrivingLicenseNumber	UNIQUE, NOT NULL	No two clients may hold the same driving licence number. This is a natural unique identifier for a client but was not chosen as the PK in favour of the more stable system-generated ClientNumber.

No other relations contain alternate keys. All other candidate keys identified during the design process coincide with the chosen primary key.

5.4 Required Data Constraints

Required data constraints (NOT NULL) specify attributes that must always hold a value. The following attributes are mandatory across all relations.

Relation	Required Attributes (NOT NULL)
OUTLET	OutletNumber, Address, PhoneNumber, FaxNumber
VEHICLE	RegistrationNumber, Model, Make, EngineSize, Capacity, CurrentMileage, DailyHireRate, OutletNumber
CLIENT	ClientNumber, FirstName, LastName, HomeAddress, PhoneNumber, DateOfBirth, DrivingLicenseNumber
HIRE_AGREEMENT	HireNumber, StartDate, EndDate, MileageBefore, ClientNumber, RegistrationNumber
STAFF	StaffNumber, FirstName, LastName, HomeAddress, HomePhoneNumber, DateOfBirth, Sex, DateJoined, JobTitle, Salary, OutletNumber

- MileageAfter in HIRE_AGREEMENT is the only nullable attribute. It is permitted to be NULL when the vehicle has not yet been returned at the time of recording.

5.5 Attribute Domain Constraints

Domain constraints define the set of permissible values for each attribute. The following table specifies domain rules for all attributes where a constraint beyond data type applies.

Attribute	Relation	Domain Constraint	Rationale
Capacity	VEHICLE	Integer ≥ 1	A vehicle must seat at least one person.
CurrentMileage	VEHICLE	Integer ≥ 0	Mileage cannot be negative.
DailyHireRate	VEHICLE	Decimal > 0.00	Hire rate must be a positive monetary value.
DateOfBirth	CLIENT / STAFF	Date < current date	Date of birth must be in the past.
Sex	STAFF	CHAR(1) IN ('M', 'F', 'O')	Constrained to M (male), F (female), or O (other).
DateJoined	STAFF	Date ≤ current date	A staff member cannot have joined in the future.
Salary	STAFF	Decimal > 0.00	Salary must be a positive monetary value.
StartDate	HIRE_AGREEMENT	Date ≤ EndDate	Hire cannot start after it ends.
EndDate	HIRE_AGREEMENT	Date ≥ StartDate	Planned return date must be on or after start date.
MileageBefore	HIRE_AGREEMENT	Integer ≥ 0	Odometer cannot be negative.

MileageAfter	HIRE_AGREEMENT	Integer ≥ MileageBefore (when not NULL)	Return mileage cannot be less than departure mileage.

5.6 General Constraints

General constraints are business rules that cannot be expressed purely through standard key or domain constraints. The following enterprise constraints apply to the Reliable Rentals schema.

#	Constraint	Applies To	Description
GC1	A vehicle cannot be on hire to more than one client simultaneously.	HIRE_AGREEMENT / VEHICLE	At any point in time, no two active hire agreements (where EndDate ≥ today and MileageAfter IS NULL) may reference the same RegistrationNumber.
GC2	A hire agreement's mileage after must be ≥ mileage before.	HIRE_AGREEMENT	When MileageAfter is recorded, it must be greater than or equal to MileageBefore, reflecting that odometers only increase.
GC3	A client must be of legal driving age.	CLIENT	The client's DateOfBirth must be at least 17 years before the StartDate of any hire agreement they are party to.
GC4	A vehicle's CurrentMileage must be updated after each completed hire.	VEHICLE / HIRE_AGREEMENT	When MileageAfter is recorded on a HIRE_AGREEMENT, the CurrentMileage on the corresponding VEHICLE record must be updated to reflect the new reading.
GC5	An outlet's FaxNumber and PhoneNumber must be distinct.	OUTLET	The fax number and phone number for an outlet should not be identical, as they represent different lines.

6. Logical ER Diagram